

Customer Lifetime Value Prediction of an Insurance Company using Regression Models

Maitri Surti

School of Technology
Pandit Deendayal Energy University
Gandhinagar, India
maitri.surti@gmail.com

Vyom Shah

School of Technology
Pandit Deendayal Energy University
Gandhinagar, India
vyom411@gmail.com

Santosh Bharti

School of Technology
Pandit Deendayal Energy University
Gandhinagar, India
Santosh.Bharti@sot.pdpu.ac.in

Rajeev Gupta

School of Technology
Pandit Deendayal Energy University
Gandhinagar, India
Rajeev.Gupta@sot.pdpu.ac.in

Abstract—Banks are faced with the issue of evaluating who would be a better investment due to the unpredictability of the economy and the unexpected changes that occur in employment. The work that has been done and is shown in the study tries to address this issue and provide an effective machine learning model that can calculate the customer lifetime value taking into consideration a variety of different criteria. Customer Lifetime Value is a statistic that assists you in determining the amount of money that you are willing to spend in order to reach and acquire clients for your company. Utilizing techniques such as exploratory data analysis (EDA) and feature selection, the primary goal of this body of work in the field of research is to find significant and deciding variables for claim submission and approval within the framework of a learning environment. In order to identify which model provides the best level of accuracy, a number of different machine-learning algorithms have been tried and tested. Therefore, the model will determine the lifetime worth of a customer by first determining the components that are the most helpful and contributing, and then taking into consideration the model that has the highest level of accuracy.

Index Terms—Customer Lifetime Value, Exploratory Data Analysis, Machine Learning, Feature Selection

I. INTRODUCTION

Machine Learning, refers to the practice of combining statistical analysis with computational research techniques. Computer engines employ this method to carry out a certain task without receiving any extra instructions. Instead, they depend on patterns that are discovered within the large data sets in order to complete the task. The field of artificial intelligence is responsible for the development of this approach to data analysis (or AI, if short). It gives the impression that computer systems are able to educate themselves and come to their conclusions with minimal input from humans. When combined with analytical forecasting, it has the potential to

make your life substantially easier. And the significance of this cannot be overstated because the calculation of lifetime value requires one to deal with a large number of numbers.

A. What is Customer Lifetime Value?

One of the most essential metrics for online retailers to track is the value of their customers across their whole lifespan. It paints a picture of the company's long-term prospects as well as its capacity to maintain its financial stability. A high Customer lifetime value is a measure of product-market fit, brand loyalty, and recurring income from current customers. If an e-commerce company wants to maintain consistent growth, it is essential that they track and maximize the worth of its customers during its whole lifespan. Customer lifetime value refers to the overall income that an e-commerce company earns from a customer over the course of its relationship with the company. It takes into consideration every order that has ever been placed by them. It is a useful indicator for gauging the contentment of customers, their loyalty, and the overall health of a business.

B. Benefits of Customer Lifetime Value

The lifetime value of a customer is significant to your company since it helps answer certain crucial issues that are used to influence future strategic decisions, including the following:

- How much do you need to invest in order to get new consumers, keep existing ones, and interact with existing ones?
- Is the expense of preserving your relationship with your consumers greater than the CLV of those customers?
- Which of your offerings—products or services—have the most potential for generating revenue?

- How will the implementation of your company strategy and marketing plans affect your bottom line?

C. Problem Statement

A car insurance business that is having problems retaining customers would decide to publicize special rates for its loyal customers as a means of attracting new ones. In 2022, the price of insurance will not be the only factor influencing the decisions of customers, and insurance companies will have to compete with a highly competitive market. They are utilizing CLV as a factor in order to categorize their most loyal clients. When this matter is resolved, the company will have access to relevant marketing strategies that can be used to increase the Customer Lifetime Value. Research of the consumer pattern inside the vehicle insurance firm is going to be conducted as part of this project. The goal of the study is to increase customer engagement and retention. It will generate a forecast about a customer's CLV while taking into account the aspects that the model has determined to be the most essential.

II. LITERATURE REVIEW

The title of the paper was "Dynamics of Customer Segments: A Predictor of Customer Lifetime Value," and the topic of this paper was the CLV of banking institutions. The purpose of this paper was to make use of the most recent data mining and big data techniques in order to create new opportunities in this area of research. The research presented in this article implies that mining patterns may be used to determine whether or not the dynamics of customer segments can be viewed as a predictor of customer lifetime value. The fact that the list of variables discussed in the literature is not exhaustive was the source of the difficulty. A problem of this study is that the data set of the target bank only contained a small portion of such characteristics, which is another restriction of the study [1].

The primary goal of this study work is to identify meaningful and decisive factors for claim filing and acceptance in a learning context through exploratory data analysis (EDA) and feature selection techniques. The paper's title was "Application of machine learning and data visualization techniques for decision support in the insurance sector," and its subtitle was "Application of machine learning and data visualization techniques for decision support in the insurance sector." In addition, the data sets have machine learning algorithms applied to them, and these algorithms are then assessed based on performance criteria. The problem with this work was that the future focus of this research might be on resolving the issue of excessive data imbalance by re-sampling data sets, grouping the abundant class, or even just adopting Boosting techniques. This could be because the problem was caused by excessive data imbalance [2].

A logistic regression model and a decision tree model were applied to actual data as a suggestion to boost the retention of consumers phoning to cancel using the CRISP-DM approach. The paper was titled "Design of a predictive model for customer retention in the cancellation of an insurance

policy," and its subtitle was "Design of a predictive model for customer retention in the cancellation of an insurance policy." When the models are compared, the results that the logistic regression model generates are superior since the accuracy of its predictions is 87.21 percent. This enables us to provide solutions that increase customer retention. Due to the fact that the article had Null entries, a gap in the study was exposed, and as a result, the entire row needed to be removed if the data being sought could not be located. In the work that has to be done in the future, we will be adding more information that is client-specific and examining how important that information is to the correctness of the models that have been deployed [3].

The title of the research paper was "Machine Learning Based Approaches to Predict Customer Churn for an Insurance Company," and the purpose of the study is to employ a variety of machine learning strategies in order to develop the most accurate model possible for predicting the rate at which customers leave an insurance company. Exploratory analysis is performed on many components of the policy in order to obtain insight into the ways in which critical parameters, such as the term of the policy and the types of coverage, impact the aim variable, which is consumers renewing or not renewing their policies. When dealing with a large data set, one of the most important challenges is finding the essential features and doing feature dimension reduction so that these characteristics may be used with a variety of potential machine learning models. It has been shown that the machine learning model with the best performance on the Area Under the Curve (AUC) metric is the Extremely Randomized Trees Classifier and Gradient Boosting Model. The issue that needed to be addressed was the fact that there are some subjects that call for more investigation. To begin, during the months of January through March and May through July, respectively, there were a significant number of contracts that were not extended. Customers' decisions to not renew their subscriptions at that time may be motivated by a variety of different factors. As a result, a superior model may be developed by including the aforementioned significant aspects [4].

The title of the study was "Feature significance analysis for User Lifetime Value prediction in games using Machine Learning: an exploratory method," and the work presented here includes a comprehensive exploratory analysis of an actual data set obtained from a large Brazilian gaming firm. The author, in collaboration with subject matter specialists from the firm, thoughtfully picked the primary aspects that would be investigated. A complete preprocessed data set that is ready to be used in a set of machine learning algorithms and a feature importance analysis based on a straightforward Random Forest Regression are the results of this analysis. Additionally, the analysis yields a set of insights that can be put into action regarding each of the selected features. Two different ways that research could be improved were found. In this study, the phase labeled "Feature Importance" is where the ranking based on impurities is implemented. To begin, feature selection that is based on impurity reduction has a natural tendency

to favor variables that have a greater number of categories. Second, when interpreting the data, it is possible to arrive at the erroneous conclusion that one of the variables is a powerful predictor while the others in the same group are unimportant when in reality, their relationships with the response variable are very similar to one another. This can occur because of the way that the data are presented [5].

The title of the article was "Does the personality of the brand's place of origin contribute to the lifetime value of retail customers? An approach to analytics based on big data ". The current study used big data analytics to analyze approximately 44,000 point-of-sale transaction records for 26,000 customers of a Taiwanese retail store in order to understand how consumer personality traits relate to country-of-origin (COO) traits (brand personality) of beer brands and to predict potential customer lifetime value. This was done in order to understand how consumer personality traits relate to the country-of-origin (COO) traits (brand personality) of beer brands (CLV). According to the findings of the research, customers usually choose and co-select businesses that have qualities that are similar to those of their personalities. It made use of a quantitative transaction data set from a single store in Taiwan, which was one of the areas that had room for development. It has been recommended that other data sets from other retailers and/or information from a wider variety of countries be used in order to get a deeper comprehension of the actions taken by consumers. Because the fact that K-means was the only method employed for clustering in the study, the conclusions cannot be generalized. As a consequence of this, it is recommended that future studies make an effort to employ Fuzzy C-means (FCM) because of the benefits it offers in terms of computing performance and clustering. This will help to expand the findings and provide alternative insights [6].

A. *Scopes of Improvement on Previous Works*

There were a few methodological flaws that were consistent across all of the studies that we read and watched, which we uncovered. To begin, the majority of the studies utilized a data set that was lacking in a sufficient number of features and/or client data, which made it impossible to get a proven CLV score that could be utilized in real-time. Second, the majority of the studies had simply updated a small number of parameters and utilized a constrained number of machine learning models. Because of this, there was less room to investigate additional methods that may improve accuracy.

III. PROPOSED WORK

The goal of the research is to both strengthen ties with existing clients and bring in new consumers. It will produce a projection regarding the lifetime value of a client while taking into account the criteria that the model has deemed are the most significant ones to take into account.

The task consists of establishing, by in-depth exploratory data analysis, the aspects that have the most significant influence on the lifetime value of a client. This includes

gaining knowledge of the link between all other aspects and the client's lifetime value, as well as identifying trends in individual features. The next step in this process is to perform some statistical analysis to evaluate whether or not the data follows a normal distribution. As a consequence of this finding, we may conclude that the Mann-Whitney U test and the Kruskal-Wallis test should be utilized to compare samples and establish whether or not a component adds to the customer's lifetime value. After this, the essential parameters are then fitted into a number of different machine learning models, and the model with the highest accuracy is chosen.

The most significant innovation brought about by this project is the fact that it is able to take into consideration the many kinds of insurance policies that each client has purchased, such as medical insurance, homeowner's insurance, and life insurance. These characteristics have been added from the outside in order to provide a more authentic picture of a situation that exists in the actual world. A more thorough study of the information is made possible by the inclusion of characteristics, and this ultimately results in the creation of a more accurate model.

IV. IMPLEMENTATION DETAILS

The primary goals of the project were to calculate the customer lifetime value and identify the elements that have the most significant influence on that value. As a result, a strategy that is comparable to that which is used in general machine learning models was chosen.

A. *Data set*

The data set that was chosen for the project includes the information of 9134 different consumers and has 27 different attributes. These characteristics include the customer's geographical location, as well as their demographic information, such as their gender and income, and even the customer's other insurance policies, which may include house insurance, life insurance, and medical insurance. The categories of data that were represented by the data entries in the data set were an object, integer, and float respectively.

B. *Preprocessing*

It was discovered that the data set in question included no instances of null values and that each of its data fields contained an entry of a valid data type. At first, the property that included the one-of-a-kind customer identification number was eliminated because it had no use in determining the worth of the client during their lifespan and was only present to separate one customer from another. After this was done, the number of distinct values that are associated with each characteristic was determined in order to assist in establishing whether or not the variables may be classified as categorical. After the preprocessing stage was over, we moved on to the exploratory data analysis phase.

C. Exploratory Data Analysis

Exploratory Data Analysis is a primary step to successfully building a machine learning model. EDA helps with a deeper comprehension of the variables in the data collection and their connections- and is usually used to investigate what data might disclose beyond the formal modeling or hypothesis testing assignment. It can also assist in determining the suitability of the statistical methods you are investigating using for data analysis. EDA's major goal is to encourage data analysis before making any assumptions. It can assist in finding glaring errors, better understanding data patterns, spotting outliers or unusual occurrences, and discovering intriguing relationships between the variables. While building the model for predicting the Customer Lifetime Value we have used uni-variate and bi-variate exploratory data analysis techniques to derive conclusions and patterns from the data set.

1) *Uni-variate Analysis*: A fundamental method of statistical data analysis is called univariate analysis. Here, there is only one variable in the data, so a cause-and-effect link is not present. Uni-variate Analysis was carried out on the dependent variable, Customer Lifetime Value to draw patterns from it by plotting distribution plots and box plots.

The distribution plot and box plot of CLV signifies that the data is highly right-skewed and there are a large number of outliers in the dependent variable. If the whisker is shorter on the lower end of the box and the median is closer to the bottom of the box, the distribution is right skewed.

For categorical variables, count plots are plotted for each feature, and from the plots, various conclusions are drawn. For example, a count plot of the state is drawn, by observing which we can conclude that a maximum number of people are residing in the state of California.

2) *Bi-variate Analysis*: A bi-variate analysis examines two variables to ascertain their correlations. In order to ascertain their empirical link, entails the investigation of two variables. The bi-variate analysis is quite useful for evaluating straightforward association hypotheses. Knowing the value of the other variable—perhaps the independent variable—can be highly useful in establishing the extent to which it is simpler to know and anticipate a value for one variable, possibly a dependent variable. Here, the bi-variate analysis was performed between the numerical independent variables and CLV (dependent variable).

D. Statistical Significance

1) *Shapiro-Wilk Test*: The Shapiro-Wilk test is a normality test that establishes whether or not the provided sample comes from a normal distribution. Shapiro's test's null hypothesis is that the population is evenly distributed. The Shapiro-Wilk Test on the CLV characteristic resulted in a p-value of zero. Since the p-value is less than 0.05, it is safe to assume that CLV data is not normally distributed.

2) *Mann - Whitney U Test*: It is a non-parametric test that is used to assess whether or not two sample means from the same population are equal by comparing their respective means. The Mann-Whitney U test is typically employed when the data is

ordinal or when the t-assumptions tests are broken. The Mann - Whitney Test is used for binary variables when the data is not normally distributed. The distributions of the two populations are identical according to the null hypothesis of this test, and they are not equal according to the alternative hypothesis H1. If the p-value is higher than 0.05, we are unable to reject the null hypothesis, and as a result, the mean rank for both binary values will be equal.

The p-values for each of the five variables included in this test—Response, Gender, Home Insurance, Medical Insurance, and Life Insurance—were all more than 0.05, so we came to the conclusion that none of these factors is statistically significant for the prediction of CLV.

3) *Kruskal - Wallis Test*: One of the non-parametric tests used as a generalized version of the Mann-Whitney U test is the Kruskal-Wallis test. It is used to test the null hypothesis, which claims that "k" samples were taken from the same population or an identical population and that the median for each sample was the same or not. According to the null hypothesis of this test, the distributions of all populations are identical, but according to the alternative hypothesis H1, they are not equal. If the p-value is higher than 0.05, we are unable to reject the null hypothesis, and as a result, the mean rank for all the values will be equal. The features which are significant according to this test are taken into consideration for model building and the rest are ignored.

E. Feature Selection

There are in total 27 features in the data set which can be utilized for predicting the Customer Lifetime Value. However, by performing various statistical tests, we could conclude that only a few of these features will make an impact on the output. To make sure that only important features are used in the model building, we perform the technique of feature selection.

The feature selection in this project is performed by making use of Recursive Backward Elimination. In this method, only the column features which had the p-value greater than 0.05 are taken into consideration while the rest of the columns are dropped off from the data set.

F. Model Selection

1) *Linear Regression*: A quiet and straightforward statistical regression technique called linear regression is utilized to highlight the link between continuous variables in predictive analysis. The term "linear regression" refers to a statistical method that displays a linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis). In linear regression, since our independent variables show a correlation with one another, there is multicollinearity in the model. To predict the results appropriately by taking into account the factor of multicollinearity, we used the Variance Inflation Factor in the linear regression model. The degree of multicollinearity in regression analysis is gauged by a variance inflation factor or VIF.

TABLE I
MODEL EVALUATION FOR LINEAR REGRESSION

RMSE	MAE	R2 Score
0.573	0.443	0.263

2) *Decision Tree Regression*: Decision tree regression trains a model in the form of a tree to predict data in the future and generate useful continuous output by observing the properties of an item. Continuous output denotes the absence of discrete output, i.e., output that is not only represented by a discrete, well-known set of numbers or values. We have fit our model using Decision Tree Regressor and have achieved satisfactory results. The results are as follows:

TABLE II
MODEL EVALUATION FOR DECISION TREE REGRESSION

RMSE	MAE	R2 Score
0.212	0.100	0.898

3) *Random Forest Regression*: As a supervised learning algorithm, Random Forest Regression does regression using the ensemble learning approach. With the use of the ensemble learning method, predictions from various machine learning algorithms can be combined to produce predictions that are more accurate than those from a single model. Several decision trees are built throughout training time in a Random Forest, and the prediction of each tree is the mean of the classes. Here, we have implemented Random Forest Regression with hyper-parameter tuning. We tuned the hyper-parameters by using Grid Search CV and found a better accuracy. The tuned hyper-parameters are ‘max depth’, ‘n estimators’, ‘bootstrap’, and ‘max features’.

TABLE III
MODEL EVALUATION FOR RANDOM FORREST REGRESSION

RMSE	MAE	R2 Score
0.211	0.103	0.899

4) *AutoML Regression*: Scikit-Learn is the foundation of the auto-sklearn AutoML system. Both open-source and cutting-edge. The designers of auto-sklearn combined strong methodologies and strategies that allowed them to take home both the first and second-place awards at the worldwide AutoML challenges. Auto-sklearn is still a very young technique overall. Many ML practitioners can immediately try auto-sklearn and observe how it functions because it is built on top of scikit-learn. The fact that this approach saves specialists a tonne of time is its most significant benefit. It is one flaw is that it functions as a “black box” and offers no guidance on how to choose. Here, ensemble learning is used, which means that various regression models are combined together to form a single model which yields better accuracy than these models individually would. For our project, AutoML is made

of Random Forest Regressor, Gradient Boosting Regressor, and AdaBoost Regressor. Of these, Gradient Boosting carried the maximum ensemble weight.

TABLE IV
MODEL EVALUATION FOR AUTOML REGRESSION

RMSE	MAE	R2 Score
0.210	0.104	0.901

V. RESULTS

The results of all three models i.e. Linear Regression, Decision Tree Regression, Random Forest Regression, and AutoML Regression were evaluated by calculating the root mean square error, mean absolute error, and R-squared score. Of the three models, the best result was provided by AutoML Regression.

TABLE V
COMPARISON OF ALL THE MODELS

Model	RMSE	MAE	R2 Score
Linear Regression	0.573	0.443	0.263
Decision Tree Regression	0.212	0.100	0.898
Random Forest Regression	0.211	0.103	0.899
Auto ML Regression	0.210	0.104	0.901

VI. SUMMARY AND FUTURE DIRECTIONS

As the world continues to progress and prices continue to rise, an increasing number of people will feel the need to purchase insurance for their vehicles if they are involved in an accident that causes damage to their vehicle. This will help them avoid incurring additional financial burdens. Since this initiative, businesses are better able to target the customers who have the most demand for their products or services and have been able to accurately anticipate the customer’s lifetime value as well as the consumer group that would benefit the most from being targeted. Future work would entail integrating more user-specific data to uncover new essential factors contributing to the customer lifetime value. This would be done to improve accuracy. The addition of more data may result in the implementation of deep learning strategies such as traditional neural networks or convolution neural networks. If you use these methods, you might be able to achieve a better level of precision while avoiding problems caused by under-fitting and over-fitting.

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